

realizableKE_BCS2

Implemented transport equations and correction terms for the realizable k-epsilon turbulence model with curvature, swirl, and optional buoyancy modifications as coded in your local OpenFOAM library.

MODEL TYPE
RAS, eddy viscosity

SOURCE FILES
realizableKE_BCS2 + BuoyantCurvatureSwirlTools2

DEFAULT CORE COEFFS
A0 = 4.0, C2 = 1.9, sigma_k = 1.0, sigma_ε = 1.2

YOUR SPLIT_RATIO_CHECK REALKE2_* CASES
Currently set to model realizableKE;

Important note: this poster describes the equations implemented in your realizableKE_BCS2 source code. The current realKE2_4000, realKE2_4500, and realKE2_5000 cases still select plain realizableKE, so these BCS2 modifications are not active there unless the model entry is changed.

1. Eddy Viscosity Closure

The model keeps the realizable k-epsilon structure and computes turbulent viscosity from a variable realizable coefficient.

TURBULENT VISCOSITY

$$\nu_t = r_{C\mu} k^2 / \epsilon$$

REALIZABLE COEFFICIENT

$$\begin{aligned} r_{C\mu} &= 1 / (A_0 + A_s U_s k / \epsilon) \\ W &= 2\sqrt{2} ((S:S):S) / (|S| S_2 + \text{small}) \\ \varphi_s &= (1/3) \cos^{-1}(\text{clamp}(\sqrt{6} W, -1, 1)) \\ A_s &= \sqrt{6} \cos(\varphi_s), \quad U_s = \sqrt{(S_2/2 + |\text{skew}(\nabla U)|^2)} \end{aligned}$$

- S = dev(symm(gradU)) in the realizable coefficient
- S2 = 2 |dev(symm(gradU))|^2 in the solver step
- ε is bounded by 0.09 k^2 / (nutMaxCoeff nu)
- omegaLike = ε / (0.09 max(k, kSmall))

2. Implemented Transport Equations

These are the equations actually assembled in realizableKE_BCS2.C.

EPSILON EQUATION

$$\begin{aligned} \partial(\alpha \epsilon) / \partial t + \nabla \cdot (\alpha \rho U \epsilon) - \nabla \cdot (\alpha \rho D_{\epsilon, \text{eff}} \nabla \epsilon) \\ = C_1 \alpha \rho |S| \epsilon - C_2 \alpha \rho \epsilon^2 / (k + \sqrt{(v \epsilon)}) - \alpha \rho C_1 B (G_b / (k + k_{\text{small}})) \epsilon + S_\epsilon \end{aligned}$$

K EQUATION

$$\begin{aligned} \partial(\alpha \rho k) / \partial t + \nabla \cdot (\alpha \rho U k) - \nabla \cdot (\alpha \rho D_{k, \text{eff}} \nabla k) \\ = \alpha \rho G f_{r, \text{eff}} - \alpha \rho (G_b / (k + k_{\text{small}})) k - (2/3) \alpha \rho k \nabla \cdot U - \alpha \rho \epsilon + S_k \end{aligned}$$

DIFFUSION AND PRODUCTION TERMS

$$\begin{aligned} D_{k, \text{eff}} &= \nu + \nu_t / \sigma_k, \quad D_{\epsilon, \text{eff}} = \nu + \nu_t / \sigma_\epsilon \\ G &= \nu_t (\nabla U : \text{dev}(2 \text{ symm}(\nabla U))) \\ \eta &= |S| k / \epsilon, \quad C_1 = \max(\eta / (5 + \eta), 0.43) \end{aligned}$$

3. BCS2 Correction Terms

The helper computes swirl gating, curvature amplification, and an optional buoyancy term.

SWIRL SENSOR

$$\begin{aligned} S_{\text{local}} &= |u_\theta| / (|U_{\text{axial}}| + \epsilon_{\text{swirl}}) \\ F_{\text{swirl}} &= \text{clamp}((S_{\text{local}} - S_0) / (S_1 - S_0), 0, 1) \end{aligned}$$

CURVATURE CORRECTION

$$\begin{aligned} r^* &= |S| / \max(|\Omega|, \omega_{\text{small}}) \\ D^2 &= \max(|S|^2, 0.09 \omega_{\text{like}}^2) \\ \tilde{r} &= (2 \text{ symm}(\Omega S) : DS/Dt) / \max(|\Omega| D^3, \text{small}) \\ f_{\text{rot}} &= (1 + c_{r1}) (2r^* / (1 + r^*)) (1 - c_{r3} \tan^{-1}(c_{r2} \tilde{r})) - c_{r1} \\ \tilde{f}_r &= \text{clamp}(f_{\text{rot}}, 0, f_{r, \text{max}}) \\ f_r &= \max(0, 1 + c_{\text{curv}}(\tilde{f}_r - 1)) \\ f_{r, \text{eff}} &= 1 + F_{\text{swirl}}(f_r - 1) \end{aligned}$$

OPTIONAL BUOYANCY TERM

$$\begin{aligned} G_b &= C_g \nu_t (g \cdot \nabla \rho) / \max(\rho, \rho_{\text{small}}) \\ B &= \tanh(|v| / u), \quad v = \hat{g} \cdot U, \quad u = |U - \hat{g} v| + u_{\text{small}} \end{aligned}$$

TYPICAL COEFFICIENTS IN YOUR DICTIONARIES

- A0 = 4.0, C2 = 1.9, sigma_k = 1.0, sigma_ε = 1.2
- S0 = 0.2, S1 = 0.8, frMax = 1.25, swirlEps = 1e-6
- Cg = 1.0, axisOrigin = (0 0 0), axisDirection = (0 0 1)
- In the current realKE2_* runs: curvature = true, swirl = true, buoyancy = false